

QUARTIC KELLER MAPS AND LEADING CURVES AT INFINITY

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ABSTRACT. The known three-dimensional counterexample to the Jacobian conjecture has ordinary coordinate degree seven, while degree at most three is classically known to be invertible. This leaves ordinary degrees four, five, and six. For $F = LX + H_2 + H_3 + H_4$, the rescaling

$$K_\epsilon(X) = \epsilon^4 F(X/\epsilon)$$

satisfies $\det JK_\epsilon = \epsilon^9 \det JF$. Thus the Keller condition is a maximal-contact condition along the determinantal hypersurface, and the leading map collapses the projective boundary to a point or a rational curve.

Our first result excludes one-dimensional leading target span: if the three coordinate forms of F_4 span a line, then a degree-at-most-four Keller map $\mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ is a polynomial automorphism. The proof reduces the transverse cubic pencil to a cube-containing or binary pencil and then uses two universal determinant syzygies.

This excludes in ordinary degree four the rank-one leading mechanism visible in the known degree-seven example. We also treat a conic-image stratum, in which

$$F_4 = G(x, y, z)(x^2, xy, y^2).$$

If the quadratic G is reduced and the base point $[0 : 0 : 1]$ is outside $G = 0$ or is a nonsingular point of it, then the Keller equations are inconsistent. The final step is an exact symbolic calculation in four parabolic orbits.

A continuation appendix closes the two proper rational-quartic frontier strata. It follows that a quartic counterexample, if one exists, has leading target span exactly two. In that locus we exclude the genuinely nonbinary quadratic-source case and prove that every coprime binary-pencil map with common ramification divisor of degree at most two is an automorphism. The surviving quartic problem is concentrated in triple-or-higher binary ramification and fixed-component boundary cases.

Working manuscript. The exact symbolic calculations have been reproduced from the recovered sources. The pencil reductions and computer-assisted conic argument still require independent specialist review.

1. INTRODUCTION

For a polynomial map $F = (F_1, F_2, F_3)$, its *ordinary degree* is

$$\deg F = \max_i \deg F_i.$$

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This convention matters: generic fiber degree and ordinary coordinate degree are different invariants.

The counterexample has ordered coordinate-degree profile $(7, 6, 4)$, and hence ordinary degree seven [Alp, Tao]. We credit Akhil Mathew as the primary human source of the problem: Mathew prompted Levent Alpöge, Alpöge prompted Fable, Fable produced the example, and Alpöge announced it publicly [Ula]. Vistoli proved that every three-dimensional Keller map of ordinary degree at most three is an automorphism [Vis99]. Thus, if $D_{\min}$ denotes the least ordinary degree of a three-dimensional counterexample, the presently known endpoints give

$$4 \leq D_{\min} \leq 7.$$

The interval is background, not a novelty claim.

Write an affine-normalized degree-four map as

$$F = LX + H_2 + H_3 + H_4, \quad L \in \mathrm{GL}_3(\mathbb{C}), \quad (1.1)$$

with H_i homogeneous of degree i . Define the *leading target span*

$$\rho_4(F) = \dim_{\mathbb{C}} \langle (H_4)_1, (H_4)_2, (H_4)_3 \rangle \subset \mathrm{Sym}^4((\mathbb{C}^3)^{\vee}). \quad (1.2)$$

This is not the generic rank of JH_4 , Waring rank, or tensor rank.

There is a complementary geometric invariant. The projectivized leading map

$$\phi_4: \mathbb{P}^2 \dashrightarrow \mathbb{P}^2, \quad [x : y : z] \mapsto [H_{4,1} : H_{4,2} : H_{4,3}]$$

has image closure $C_4(F)$. The determinant equation implies that this image has dimension at most one. Subject to base-locus degeneracies, the cases $\rho_4 = 1, 2, 3$ correspond to point, line, and nondegenerate rational-curve image. We use ρ_4 for exact linear algebra and $C_4(F)$ to organize the geometry at infinity.

Our main theorem excludes $\rho_4(F) = 1$. The continuation in appendix A then excludes the remaining proper nondegenerate rational leading curves and begins the resulting target-span-two endgame. The further continuation in appendix B closes the primitive ramification-four/five strata and the fixed-component leaves, conditional on the recorded upstream routing.

Theorem 1.1 (One-dimensional leading target span). *Let $F: \mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ be a Keller map of ordinary degree at most four. If $\rho_4(F) = 1$, then F is a polynomial automorphism.*

The theorem allows arbitrary mixed quadratic and cubic layers. It assumes neither that $F - X$ is homogeneous nor that its Jacobian is nilpotent. This distinguishes it from the homogeneous three-dimensional theorem of de Bondt–van den Essen [dBvdE05]. General invariant theory of binary quartic pencils is classical [Wal98]; our claim concerns the restrictions imposed by the mixed homogeneous Keller equations.

Recent work restricts graded Keller maps [Sha26]. We impose no torus action. As of July 23, 2026, we know no public theorem excluding every ordinary-degree-four three-dimensional Keller map.

Our second theorem concerns a conic-image leading form rather than linear target span two.

Theorem 1.2 (Four reduced conic orbits). *Suppose a degree-four Keller map has, after independent linear source and target changes,*

$$H_4 = G(x, y, z)(x^2, xy, y^2),$$

where G is a reduced quadratic form. Put $p = [0 : 0 : 1]$. If $p \notin V(G)$, or if p is a nonsingular point of $V(G)$, then no such Keller map exists.

The final coefficient elimination in theorem 1.2 is computer-assisted. The calculation keeps an arbitrary invertible linear part and never divides by a parameter. The exact scope is described in section 9.

2. KELLER JETS AT INFINITY

The homogeneous determinant identities have a compact geometric source. Define

$$K_\epsilon(X) = \epsilon^4 F(X/\epsilon) = H_4 + \epsilon H_3 + \epsilon^2 H_2 + \epsilon^3 L X.$$

Since $JK_\epsilon = \epsilon^3 JF(X/\epsilon)$,

$$\det JK_\epsilon = \epsilon^9 \det JF. \quad (2.1)$$

In particular, $\det JH_4 = 0$. More strongly, every coefficient below the maximum possible order nine in the determinant arc

$$JH_4 + \epsilon JH_3 + \epsilon^2 JH_2 + \epsilon^3 L$$

vanishes. Thus a quartic Keller map is a maximally volume-preserving deformation of a singular leading map at infinity.

The rank-one theorem treats the case in which $C_4(F)$ is a point. The span-two reduction treats line image. The conic theorem begins the first nonlinear case. This explains why the cases belong in one paper: they are successive image geometries of the same boundary jet, rather than unrelated coefficient ansatzes.

Remark 2.1 (Determinantal arcs). Equation (2.1) says that the matrix arc JK_ϵ has maximal contact with the determinant hypersurface in $\text{Mat}_{3 \times 3}$. Our matrices are not arbitrary arcs: each coefficient is the Jacobian of a homogeneous polynomial map. The integrability constraint is what turns a coarse determinantal problem into the pencil, coordinate, and first-integral geometry used below.

3. TWO COORDINATE LEMMAS

We work temporarily over an algebraically closed field k of characteristic zero.

Lemma 3.1 (Quadratic coordinate). *If $f \in k[X_1, X_2, X_3]$ has degree at most two and no critical point, then some constant directional derivative of f is nonzero. In particular, f is a coordinate.*

Remark 3.2 (Sharpness of low-degree coordinate rigidity). The quadratic hypothesis is essential. The polynomial

$$f = x + x^2y$$

has no critical point, but its zero fiber is reducible, so it is not a coordinate. Likewise $x^4 + x + x^2y$ has pure-power quartic top part, no critical point, and a reducible zero fiber. The coordinate lemmas therefore encode a genuine low-degree threshold rather than an unrestricted submersion principle.

Proof. Write

$$f = \frac{1}{2}X^TAX + b^TX + c$$

with A symmetric. If $b \in \text{im } A$, the equation $AX + b = 0$ has a solution, contrary to the hypothesis. Hence

$$b \notin \text{im } A = (\ker A)^\perp.$$

There is $u \in \ker A$ with $b^Tu \neq 0$, and $D_u f = b^Tu$. After making u a coordinate direction, a triangular change replaces that coordinate by f . $\square$

Lemma 3.3 (Cubic-cube coordinate). *Suppose that $\deg f \leq 3$, f has no critical point, and its cubic homogeneous part is $\lambda\ell^3$ with $\lambda \neq 0$. Then f has a nonzero constant directional derivative and is a coordinate. The straightening and its inverse can be chosen of degree at most three.*

Proof. Take $\ell = x$, write the remaining variables as $w \in k^2$, and express

$$\nabla_w f = Mw + ax + b$$

with M symmetric. Project this affine expression to $\text{coker } M = (\ker M)^\vee$.

If it never vanishes, the two projected vectors a, b do not span an affine line through the origin. Hence some $u \in \ker M$ satisfies $u^Ta = 0$ and $u^Tb \neq 0$, so $D_u f$ is a nonzero constant.

Otherwise fix a solution of $Mw + ax + b = 0$. If a pairs nontrivially with $\ker M$, variation within the solution space makes f_x vanish. If it does not, then $a \in \text{im } M$; solvability for one x also puts b in $\text{im } M$, so the system is solvable for every x . Substitution of a linear solution $w = w(x)$ makes f_x a quadratic polynomial in x with leading coefficient 3λ . It has a root over k , again producing a critical point. Both alternatives contradict the hypothesis, so the constant-direction case is forced.

In suitable linear coordinates $f = cw_2 + \psi(x, w_1)$, with $c \neq 0$ and $\deg \psi \leq 3$. Replacing w_2 by f gives the asserted straightening and degree bound. $\square$

Proposition 3.4 (The easy transverse branches). *In (1.1), suppose*

$$H_4 = (0, 0, h), \quad H_3 = (P, Q, R).$$

If P, Q are linearly dependent, or if their pencil contains the cube of a linear form, then F is an automorphism.

Proof. A row of JF cannot vanish, so every target combination of the coordinates of F has no critical point. In the dependent case a target combination of the first two components has degree at most two, and lemma 3.1 applies. In the cube-containing case a combination has degree at most three with cubic part a cube, and lemma 3.3 applies.

After straightening this component to a variable t , choose the other target combination among the first two original components. Over $\overline{k(t)}$, the remaining two coordinates form a plane Keller map. In the quadratic case one component has degree at most six. In the cubic case it has degree in

$$\{1, 2, 3, 4, 5, 6, 7, 9\}.$$

Indeed, substituting $w_2 = c^{-1}(t - \psi)$ into a cubic gives degree at most seven unless the w_2^3 term and the cubic part of ψ are both nonzero, in which case the unique top term has degree nine. Every displayed integer is a product of at most two primes, counted with multiplicity. The corrected Appelgate-Onishi plane theorem, in the form completed by Nagata, therefore makes the plane map an automorphism [AO85, Nag88]. Its inverse over $\overline{k(t)}$ is unique, hence fixed by every automorphism over $k(t)$, and therefore descends to $k(t)$. It follows that F is birational over k . A birational Keller self-map is injective by Zariski's Main Theorem and hence an automorphism by Ax-Grothendieck. $\square$

4. THE RATIONAL-COMPOSITE PENCIL LEMMA

The highest nonautomatic homogeneous Keller equation for the normalization in proposition 3.4 is

$$dP \wedge dQ \wedge dh = 0. \quad (4.1)$$

If P, Q are independent, they are algebraically independent and (4.1) makes h algebraic over $k(P, Q)$.

Lemma 4.1 (Weighted one-variable field). *Let P, Q be homogeneous forms of the same degree $a > 0$. Write*

$$P = GA, \quad Q = GB, \quad \gcd(A, B) = 1, \quad n = \deg A = \deg B,$$

and let h be homogeneous of degree $b > 0$. Put

$$\begin{aligned} t &= P/Q = A/B, & g &= \gcd(a, b), \\ p &= a/g, & q &= b/g, & w &= h^p/Q^q, \end{aligned}$$

and suppose that h is algebraic over $k(P, Q)$. Then w is algebraic over $k(t)$.

If $K_1 = k(t, w)$, there are coprime homogeneous forms A_0, B_0 of a common degree d and a rational function $\mathcal{R}$ of degree e such that

$$K_1 = k(A_0/B_0), \quad t = \mathcal{R}(A_0/B_0), \quad n = ed.$$

Proof. Choose a polynomial relation among P, Q, h and retain a nonzero source-homogeneous part. Every monomial $P^i Q^j h^k$ in that part satisfies

$$a(i + j) + bk = M.$$

It follows that all exponents k are congruent modulo $p = a/g$. After removing their common residual power of h , the relation has the form

$$\sum_i C_i(P, Q)h^{pi} = 0, \quad \deg C_i = N - qi$$

for a suitable integer N . Division by Q^N gives a polynomial relation for w over $k(t)$.

The one-variable field $K_1 \subset k(\mathbb{P}^2)$ is the function field of a curve dominated by $\mathbb{P}^2$. Restriction to a suitable line makes that curve unirational, hence rational in characteristic zero. Thus $K_1 = k(s)$, and s can be written A_0/B_0 with A_0, B_0 coprime homogeneous forms of the same degree d .

Write $\mathcal{R} = U/V$, where U, V are coprime binary forms of degree e . The substituted forms $U(A_0, B_0)$ and $V(A_0, B_0)$ remain coprime: their common divisor would, by the binary resultant identities, divide powers of both A_0 and B_0 . Comparing degrees in the reduced ratio $A/B = \mathcal{R}(A_0/B_0)$ yields $n = ed$. $\square$

Proposition 4.2 (Cubic pencils). *Let P, Q be independent homogeneous cubics satisfying (4.1). Then their pencil either contains the cube of a linear form, or P, Q, h are polynomials in two independent linear forms.*

Proof. First suppose $\gcd(P, Q) = 1$. Apply lemma 4.1 directly to P, Q, h . If $e > 1$, then $3 = ed$ gives $e = 3, d = 1$, so P, Q are binary in two linear forms.

If $e = 1$, write $w = R(t)$. For $F_\xi = P - \xi Q$, let $r_\xi = \text{ord}_\xi R$, and set

$$c_\xi = r_\xi + 4\mathbf{1}_{\xi=\infty}.$$

If an irreducible component of F_ξ has multiplicity m , valuation gives

$$3\nu(h) = c_\xi m.$$

Every c_ξ is nonnegative and

$$\sum_{\xi \in \mathbb{P}^1} c_\xi = 4.$$

If no fiber is a cube, every cubic fiber has a component whose multiplicity is prime to three. The displayed valuation then makes every c_ξ a multiple of three, contradicting their sum.

Now write $P = GA, Q = GB$ with $G \neq 1$ and $\gcd(A, B) = 1$. If $\deg G = 2$, then A, B are independent linear forms. An irreducible factor of G , of multiplicity $m = 1$ or 2 , cannot carry a nonconstant A/B , since its valuation would give

$$3\nu(h) = 4m.$$

Every component of G is therefore a fiber line of the pencil $\langle A, B \rangle$; hence G, P, Q are binary.

If $G = \ell$, the reduced pencil A/B has degree two. In the composite case lemma 4.1 gives $e = 2, d = 1$. The valuation along ℓ forces ℓ into the same binary pencil. In the primitive case, the same valuation first makes A/B constant on $\ell = 0$. After changing the pencil basis, write $A = \ell m$. If m

is proportional to ℓ , then $P = \ell^3$. Otherwise, with $r = \text{ord}_0 R$, valuations along $m = 0$ and $\ell = 0$ give

$$3\nu_m(h) = r, \quad 3\nu_\ell(h) = r + 4,$$

which is impossible modulo three.

Finally, if P, Q are binary in linear forms a, b , then

$$dP \wedge dQ = J_{a,b}(P, Q) da \wedge db$$

with nonzero planar Jacobian. Equation (4.1) forces the derivative of h transverse to a, b to vanish, so h is binary in the same forms. $\square$

5. PROOF OF THE ONE-DIMENSIONAL-SPAN THEOREM

By a target change, write $H_4 = (0, 0, h)$. In the branch not already covered by proposition 3.4, proposition 4.2 gives coordinates such that

$$P, Q \in k[x, y]_3, \quad h \in k[x, y]_4,$$

and the pencil $\langle P, Q \rangle$ contains no cube.

Put

$$W = J(P, Q), \quad U = J(P, h), \quad V = J(Q, h),$$

where J is the planar Jacobian in x, y . Write every possible z -dependent lower term as

$$\begin{aligned} (H_2)_1 &= A_0(x, y) + za(x, y) + \alpha z^2, \\ (H_2)_2 &= B_0(x, y) + zb(x, y) + \beta z^2, \\ (H_2)_3 &= C_0(x, y) + zr(x, y) + cz^2, \\ (H_3)_3 &= R_0(x, y) + zq(x, y) + z^2\ell(x, y) + \rho z^3, \end{aligned} \tag{5.1}$$

where a, b, r, ℓ are linear and q is quadratic.

Introduce T and expand

$$\det(L + TJH_2 + T^2JH_3 + T^3JH_4) = \det L + \sum_{j=1}^7 T^j E_j.$$

Exact expansion gives

$$[z^2]E_6 = 3\rho W, \tag{5.2}$$

$$[z]E_6 = 2(W\ell + \alpha V - \beta U), \tag{5.3}$$

$$[1]E_6 = Wq + Va - Ub, \tag{5.4}$$

$$[z^3]E_5 = -2J(\ell, \alpha Q - \beta P), \tag{5.5}$$

where the last three identities are read after $\rho = 0$. After $\alpha = \beta = \ell = \rho = 0$, one also has

$$\begin{aligned} [z]E_5 = & -4J(a, b)h - qJ(a, Q) + 3QJ(a, q) \\ & + qJ(b, P) - 3PJ(b, q) + 2cW, \end{aligned} \quad (5.6)$$

$$[z^2]E_4 = -J(a, b)q + 2c(J(a, Q) - J(b, P)), \quad (5.7)$$

$$[z^3]E_3 = 2cJ(a, b). \quad (5.8)$$

Lemma 5.1 (First binary syzygy). *In the cube-free branch,*

$$\rho = \ell = \alpha = \beta = 0.$$

Proof. Equation (5.2) gives $\rho = 0$. Put $S = \alpha Q - \beta P$. Equations (5.3) and (5.5) say

$$W\ell + J(S, h) = 0, \quad J(\ell, S) = 0.$$

If $\ell \neq 0$, the second equation makes $S = \lambda\ell^3$. Cube-freeness gives $S = 0$, whence $\alpha = \beta = 0$, and the first equation becomes $W\ell = 0$, a contradiction.

Thus $\ell = 0$, and $J(S, h) = 0$. Algebraically dependent homogeneous binary forms of coprime degrees three and four are powers of a common linear form. A nonzero S would therefore be a cube in $\langle P, Q \rangle$. Hence $S = 0$, and independence of P, Q gives $\alpha = \beta = 0$. $\square$

Lemma 5.2 (Second binary syzygy). *In the cube-free branch,*

$$a = b = q = c = 0.$$

Proof. Let $D = J(a, b)$. If $D \neq 0$, equations (5.8), (5.7), and (5.6), in that order, give $c = 0, q = 0, h = 0$, a contradiction. Thus $D = 0$.

If $a = b = 0$, equations (5.4) and (5.6) give $q = c = 0$. Otherwise a row operation on the first two target coordinates makes $b = 0, a \neq 0$. If $c \neq 0$, (5.7) makes $J(a, Q) = 0$, so Q is a cube. Hence $c = 0$.

Equation (5.6) is now

$$-qJ(a, Q) + 3QJ(a, q) = 0.$$

If $q = 0$, equation (5.4) gives $J(Q, h) = 0$, again making Q a cube. Otherwise take $a = x$; then

$$qQ_y - 3Qq_y = 0, \quad \partial_y(Q/q^3) = 0.$$

Homogeneity gives $x^3Q = \lambda q^3$. Unique factorization makes Q a cube, the final contradiction. $\square$

Proof of theorem 1.1. The two syzygy lemmas reduce the map to

$$F_1 = f_1(x, y) + \gamma z, \quad F_2 = f_2(x, y) + \phi z, \quad F_3 = f_3(x, y) + (\mu + r(x, y))z,$$

with r linear. The coefficient of z in the determinant is

$$J(r, \phi f_1 - \gamma f_2).$$

If $(\gamma, \phi) \neq (0, 0)$ and $r \neq 0$, its vanishing makes the nonzero cubic $\phi P - \gamma Q$ a polynomial in the single linear form r , hence a cube. Therefore $r = 0$. A

target linear change sends the nonzero vector (γ, ϕ, μ) to e_3 , and the map becomes

$$(G_1(x, y), G_2(x, y), z + G_3(x, y)).$$

The plane pair is Keller and one coordinate has degree at most four, so [AO85] makes it an automorphism.

If $\gamma = \phi = 0$, then

$$\det JF = J(f_1, f_2)(\mu + r).$$

Both factors are units, so $r = 0$, and the same triangular conclusion holds. $\square$

6. A LEADING-TARGET-SPAN-TWO REDUCTION

For context, suppose instead that

$$H_4 = (P, Q, 0)$$

with P, Q algebraically independent quartics, and let R be the third cubic component of H_3 . The highest Keller equation is

$$\text{Jac}(P, Q, R) = 0.$$

Write $P = GA, Q = GB$, with $\gcd(A, B) = 1$, and set

$$n = \deg A = \deg B = 4 - \deg G, \quad t = A/B, \quad w = R^4/Q^3.$$

The proof of lemma 4.1, with weights 4, 4, 3, gives a rational intermediate field and a factorization $n = ed$.

Here and below, a reduced pencil A/B is *composition-primitive* if this factorization has $e = 1$. This is distinct both from coprimality of A, B and from the period-primitivity used for homogeneous invariant fields.

Proposition 6.1 (Four structural loci). *Every such leading pair belongs to at least one of the following loci:*

- (i) a binary quartic pencil, with R binary in the same two linear forms;
- (ii) a quadratic-source pencil

$$P = U(a_2, b_2), \quad Q = V(a_2, b_2),$$

where a_2, b_2 are coprime quadrics and U, V are binary quadratics;

- (iii) a composition-primitive coprime pencil containing a fourth power ℓ^4 ;
- (iv) a composition-primitive reduced pencil A/B with a nontrivial common factor G , every irreducible component of G not already binary being supported on a special fiber of A/B .

The loci need not be disjoint.

Proof. If the intermediate extension has degree $e > 1$, the possibilities $n = ed$ are

$\deg G$	n	(e, d)
0	4	$(4, 1), (2, 2)$
1	3	$(3, 1)$
2	2	$(2, 1)$.

The $d = 1$ cases are binary after valuation along every component of G ; the only nonbinary composite case is $(e, d) = (2, 2)$.

In the composition-primitive coprime case $w = \rho(t)$. If a component of multiplicity m lies in the fiber $P - \xi Q$, then

$$4\nu(R) = c_\xi m, \quad c_\xi \geq 0, \quad \sum_{\xi \in \mathbb{P}^1} c_\xi = 3,$$

with the denominator contribution incorporated at infinity. Some c_ξ is odd, so every component multiplicity in that fiber is divisible by four. The quartic fiber is therefore ℓ^4 . With fixed components, the same valuation argument places every component not already binary over a special fiber, giving (iv). $\square$

Remark 6.2. Proposition 6.1 is a structural reduction, not an exclusion theorem. Its relation to the classical stratification of binary quartic pencils [Wal98] is that the Keller equation supplies the cubic R and the valuation restrictions; it does not make the ambient pencil theory new.

7. THE FOUR REDUCED CONIC ORBITS

We now turn to the separate conic-image form

$$H_4 = G(x, y, z)(x^2, xy, y^2). \quad (7.1)$$

Put $p = [0 : 0 : 1]$.

Lemma 7.1 (Parabolic orbit list). *Up to a source transformation fixing p and rescaling G , the seven orbits of pairs consisting of a plane conic and p have representatives*

<i>geometry</i>	<i>G</i>
<i>smooth, $p \notin V(G)$</i>	$z^2 + xy$
<i>smooth, $p \in V(G)$</i>	$xz + y^2$
<i>two lines, $p \notin V(G)$</i>	$z^2 + x^2$
<i>two lines, p nonsingular on the union</i>	xz
<i>two lines, p the node</i>	xy
<i>double line, $p \notin V(G)$</i>	z^2
<i>double line, $p \in V(G)$</i>	x^2 .

Proof. The subgroup preserving the pencil $\langle x, y \rangle$ is the projective stabilizer of p . Over $\mathbb{C}$, plane conics are classified by rank. For rank three, the stabilizer is transitive on points on the conic and on points off it. For rank two, a point is off the union, on exactly one component, or at the node. For rank one, it is on or off the doubled line. Elementary linear changes give the displayed representatives. $\square$

For (7.1), define

$$\delta_G = xG_z\partial_x + yG_z\partial_y + (zG_z - 4G)\partial_z.$$

This is a right-kernel derivation for JH_4 . Let

$$n = (y^2, -2xy, x^2)^T.$$

Then n is a left-kernel vector, and

$$\delta_G(n) = 2G_z n.$$

Lemma 7.2 (Invariant fields). *For each of*

$$G = z^2 + xy, \quad xz + y^2, \quad z^2 + x^2, \quad xz,$$

one has

$$\ker_{\mathbb{C}(x,y,z)} \delta_G = \mathbb{C} \left(\frac{x}{y}, Gy^2 \right).$$

Consequently a homogeneous invariant rational function has scalar degree divisible by four.

Proof. Put $r = x/y$ and $s = Gy^2$. Direct differentiation gives $\delta_G(r) = \delta_G(s) = 0$.

For $G = xz + \epsilon y^2$, the full field is $\mathbb{C}(r, s)(y)$, since

$$z = \frac{s - \epsilon y^4}{ry^3}.$$

The induced nonzero derivation on this one-variable rational field has constant field $\mathbb{C}(r, s)$.

For $G = z^2 + xy$ and $z^2 + x^2$, put $w = yz$. Over $\mathbb{C}(r, s)$, the full field is the function field of

$$w^2 + ry^4 = s \quad \text{or} \quad w^2 + r^2 y^4 = s.$$

Each generic curve is geometrically integral. Thus $\mathbb{C}(r, s)$ is relatively algebraically closed in the one-variable function field, and a nonzero $\mathbb{C}(r, s)$ -derivation has no further constants.

Under scalar dilation, r has degree zero and s degree four. The rational eigenvectors in $\mathbb{C}(r, s)$ therefore have degrees in $4\mathbb{Z}$. $\square$

Write $H_3 = (P, Q, R)$ and $M = n \cdot H_3$. The first normal Keller equation is

$$n^T JH_3 \delta_G = 0,$$

or

$$\delta_G(M) = 2G_z M.$$

Hence M/y^2 is invariant of degree three. By lemma 7.2, $M = 0$, and the syzygies of $(y^2, -2xy, x^2)$ give

$$H_3 = T(A, B) := (2xA, yA + xB, 2yB) \tag{7.2}$$

for quadratic forms A, B .

Introduce

$$K_\epsilon = H_4 + \epsilon H_3 + \epsilon^2 H_2 + \epsilon^3 LX, \quad \Phi(Y) = Y_1 Y_3 - Y_2^2.$$

The chain rule gives

$$\text{Jac}(K_{\epsilon,1}, K_{\epsilon,2}, \Phi(K_{\epsilon})) = K_{\epsilon,1} \det JK_{\epsilon}. \quad (7.3)$$

For a Keller map,

$$\det JK_{\epsilon} = \epsilon^9 \det L.$$

Write

$$\Phi(K_{\epsilon}) = \epsilon^2 S_2 + \epsilon^3 S_3 + \epsilon^4 S_4 + \epsilon^5 S_5 + \epsilon^6 S_6.$$

The first two coefficients of (7.3) make S_2, S_3 δ_G -invariant. Their degrees are six and five, so lemma 7.2 gives

$$S_2 = S_3 = 0.$$

Now

$$S_2 = G n \cdot H_2 - (yA - xB)^2.$$

Since G is squarefree,

$$yA - xB = G\ell, \quad n \cdot H_2 = G\ell^2 \quad (7.4)$$

for a linear form ℓ . In fact $\ell \in (x, y)$. For $G = z^2 + xy$ or $z^2 + x^2$, reduce the first identity modulo (x, y) : since $yA - xB \in (x, y)$, one obtains $z^2 \bar{\ell} = 0$ in $k[z]$. For $G = xz + y^2$ or xz , use the second identity: $n \cdot H_2 \in (x, y)^2$, whereas G has (x, y) -order one, so $\ell \notin (x, y)$ would make $G\ell^2$ have order one. Thus in every case $\ell = ax + by$. Every solution of the second equation is

$$H_2 = G(b^2, -ab, a^2) + T(U, V),$$

where U, V are linear. The first equation has the exhaustive solutions

G	A	B
$z^2 + xy$	$xC + ax^2 + bz^2$	$yC - az^2 - by^2$
$xz + y^2$	$xC + axy + bG$	$yC - axz$
$z^2 + x^2$	$xC + bG$	$yC - aG$
xz	$xC + bG$	$yC - aG,$

with C linear.

Proposition 7.3 (Exact terminal elimination). *For each of the four displayed G , the equations*

$$S_3 = 0, \quad \delta_G(S_4) = 0$$

and one further coefficient of (7.3) force $\det L = 0$.

Computer-assisted proof. Substitute the preceding parametrizations and retain all nine entries of L . Solving the linear equation $S_3 = 0$ leaves two entries of L free. The coefficients of $\delta_G(S_4) = 0$ are then solved without inverting a parameter. The orbit $G = xz + y^2$ has the exhaustive split

$$a = 0 \quad \text{or} \quad [z]C = [x]C.$$

In the resulting five charts, the next nonzero coefficient and the determinant have the form

G	required zero coefficient	$\det L$
$z^2 + xy$	$x^2 z G(A_0 x + 2B_0 y)^2$	$c_2(-bA_0 + 2aB_0)^2$
$xz + y^2, c_2 = c_0$	$\frac{1}{2}x^3 G(A_0 x - 2my)^2$	$c_2(bA_0 + 2am)^2$
$xz + y^2, a = 0$	$\frac{1}{2}x^3 G(nx + (4b^2(c_2 - c_0) - 2m)y)^2$	$b^2 c_2 n^2$
$z^2 + x^2$	$x^2 z G(A_0 x + B_0 y)^2$	$c_2(bA_0 - aB_0)^2$
xz	$\frac{1}{2}x^3 G(A_0 x + B_0 y)^2$	$c_2(bA_0 - aB_0)^2$

Here the displayed letters are the residual scalar parameters after the linear solves. In every row, vanishing of the middle polynomial forces the squared linear form to vanish coefficientwise, and the last column then vanishes.

The accompanying assertion-based checker constructs the parametrizations, performs the linear solves, verifies the exhaustive branch equation, expands (7.3), and asserts each factorization exactly over $\mathbb{Q}$. The evidence boundary is stated in section 9. $\square$

Proof of theorem 1.2. Lemma 7.1 reduces the hypothesis to the four forms treated in proposition 7.3. That proposition contradicts the required invertibility of L . $\square$

Remark 7.4 (The three untreated conic orbits). The remaining representatives are

$$G = xy, \quad G = z^2, \quad G = x^2.$$

For $G = xy$ and $G = x^2$, the characteristic derivation has odd-degree polynomial invariants, so the first normal deformation need not vanish. For $G = z^2$, degree-six invariants survive, so S_2 need not vanish. The proof above does not specialize to these cases.

8. WHY THE DEGREE-SEVEN EXAMPLE ESCAPES

The public degree-seven map has a rank-one top layer, so it is important to say exactly why theorem 1.1 does not continue formally to degree seven. In the displayed coordinates, one transverse component is

$$R = 2x - 3x^2y - x^3z = x(2 - 3xy - x^2z).$$

It is a polynomial submersion:

$$R_y = -3x^2, \quad R_z = -x^3,$$

so a critical point would have $x = 0$, where $R_x = 2$. Nevertheless R is not a coordinate, because the fiber $R^{-1}(0)$ is reducible.

In ordinary degree four, the dependent transverse branch in the rank-one argument produces a quadratic polynomial submersion, and lemma 3.1 forces it to be a coordinate. The known counterexample reaches a quartic submersion that is smooth but globally nontrivial. Thus the degree gap is not an artifact of the proof: low-degree differential regularity still forces algebraic triviality, whereas degree seven has enough room for nontrivial behavior at infinity.

9. EXACT COMPUTATIONS AND REMAINING PROOF OBLIGATIONS

The recovered supplement contains one universal rank-one script and nine conic-orbit scripts. All ten programs were rerun with Python 3.12.11 and SymPy 1.14.0. Every new log is byte-for-byte identical to the stored output. The reviewed copies, stored outputs, and integrity manifests are in `code/rank-one/` and `code/recovered-conic/`.

For the conic theorem, the standalone checker

`code/verify_conic_certificates.py`

reconstructs the five terminal charts from the normal equations. It asserts the cubic and quadratic normal forms, $S_2 = S_3 = 0$, $\delta_G(S_4) = 0$, the set-theoretically exhaustive split $a = 0$ or $[z]C = [x]C$, and all five pairs of factorizations in proposition 7.3. It uses exact rational arithmetic, makes no random specializations, and divides by no parameter.

The rank-one script keeps arbitrary binary cubics P, Q , an arbitrary binary quartic h , every lower term in (5.1), and an arbitrary 3×3 linear part. It asserts (5.2)–(5.8) and the terminal determinant identity. It does not prove lemmas 3.3 and 4.1 and proposition 4.2.

The bounded equivariant calculation has a separate source directory. Its core scripts recompute the degree-four, degree-five, and degree-six eliminations over $\mathbb{Q}$, together with the exact target-orbit minor. Exploratory finite-field searches are kept in a visibly separate subdirectory and are not used as characteristic-zero nonexistence proofs. The computation index records which script supports each computer-assisted step.

The continuation supplement

`code/program-2-2026-07-27-v1/`

contains five proof bundles and eight exact checkers. They cover the rational-quartic frontier, the fixed-component Plücker chart, regular binary pencils, nine quadratic-source charts, simple ramification, and degree-two ramification. All eight scripts were source-reviewed and rerun successfully. The rational-quartic and Plücker calculations each have a second implementation using only standard-library sparse polynomial arithmetic. The file `SHA256SUMS` pins every imported source.

The later degree-three F_4 calculation has no attached complete checker. Accordingly, appendix A records the proved generic reduction and the remaining compatibility equation, but does not state a complete degree-three theorem.

The July 29 continuation supplement

`code/program-2-2026-07-29-v2/`

contains the proof packets for high ramification, zero minors, fourth powers, and fixed components. Every standalone exact program was reviewed and replayed. The quadratic fixed-factor archive matched its internal hash manifest, and all 38 replay groups passed from a fresh copy. These replays

certify the encoded identities and finite charts. They do not certify the leading-curve/four-loci case tree or map every proof chart to a program.

The checker makes the computational implication in proposition 7.3 explicit. It does not certify lemmas 7.1 and 7.2; those are conventional arguments in the manuscript. A submission release therefore requires:

- (1) an independent audit of the rational-composite-pencil argument;
- (2) a second check of the parabolic orbit list and invariant fields; and
- (3) specialist review of the computer-assisted implication in proposition 7.3.

10. OPEN DEGREE-FOUR STRATA

Corollary A.2 shows that an ordinary-degree-four counterexample must have leading target span exactly two. Theorem A.4 eliminates the genuinely nonbinary quadratic-source locus and the coprime binary locus through two units of common ramification. Subject to the upstream routing, theorems B.1, B.4 and B.5 also close ramification degrees four and five and all fixed-factor degrees. These results do not settle ordinary degree four. The remaining work is:

- the explicit exceptional degree-three weighted-inflection compatibility problem described at the end of appendix A; and
- a global audit that the leading-curve and four-loci reductions route every quartic map into the recorded terminal charts.

The natural next calculation is the extension-field D_6 -to- D_5 compatibility check on the remaining F_4 family. Even if that calculation closes, the global audit and independent proof review remain necessary before claiming $D_{\min} \geq 5$.

APPENDIX A. CLOSING THE RATIONAL-QUARTIC FRONTIER

This appendix records a continuation of the leading-curve analysis. It has two purposes. First, it closes the two proper rational-quartic strata left by the frontier theorem. Second, once the leading target span has thereby been reduced to two, it eliminates the regular, simple-ramification, and double-ramification parts of the coprime binary-pencil locus, together with the genuinely nonbinary quadratic-source locus. The complete proof notes and exact checkers are preserved in

`code/program-2-2026-07-27-v1/`.

A.1. The two proper rational-quartic strata. Let

$$h = (h_0, h_1, h_2) \in \mathbb{C}[x, y]_4^3$$

be a basepoint-free proper parametrization of a plane quartic. Write

$$h_x \times h_y = q n$$

with n primitive. If

$$H_3 = C_3(x, y) + zV(x, y), \quad H_2 = D_2(x, y) + zW_1(x, y) + z^2u,$$

the first two determinant-jet equations are

$$(h_x \times h_y) \cdot V = 0, \tag{A.1}$$

$$[z]E_2 = Q_h(V) + 2(h_x \times h_y) \cdot u = 0, \tag{A.2}$$

where

$$Q_h(V) = \det(V_x, h_y, V) + \det(h_x, V_y, V).$$

If $V = 0$, the same equations make $W_1 + 2zu$ a low-degree syzygy of n . If both lower nonlinear layers are binary, the next equation is

$$E_3 = (h_x \times h_y) \cdot L_z,$$

where L_z is the third column of L .

Theorem A.1 (Rational-quartic frontier exclusion). *Assume the preceding frontier reduction: every proper rational-quartic leading image not already excluded has either*

$$(\deg q, \text{ tangent-syzygy type}) = (3, (1, 2)) \quad \text{or} \quad (2, (2, 2)).$$

Neither stratum occurs as the projective leading image of a quartic Keller map.

Proof. For type $(3, (1, 2))$, use the tricuspidal parametrization

$$h = (x^2y^2, y^2(x - y)^2, x^2(x - y)^2).$$

Its derivative has the factorization

$$h_x \times h_y = 8xy(x - y)n, \quad n = (-(x - y)^3, x^3, -y^3),$$

and a Hilbert–Burch basis

$$a = (x + y, x - 2y, -2x + y), \quad b = (x^2, x^2 - 3xy + 3y^2, x^2).$$

Thus the most general first normal cubic layer is

$$z((px + qy)a + \mu b) + \rho z^2a.$$

The z^3 -coefficient of the second jet is

$$24\rho^2(x^2 - xy + y^2)^2,$$

so $\rho = 0$. The next coefficient is

$$12\kappa^2 + 16xy(x - y)(n \cdot u),$$

where

$$\kappa = (\mu + p)x^3 + (-2\mu - p + q)x^2y + (p - q)xy^2 + qy^3.$$

Divisibility by the squarefree cubic $xy(x - y)$ forces $\kappa = cxy(x - y)$; comparison of coefficients gives

$$p = q = \mu = c = 0.$$

The remaining low-degree syzygy is a scalar multiple of a , and a later coefficient is $12\sigma^2(x^2 - xy + y^2)^2$. Hence the quadratic layer is binary as well. The terminal equation then gives $L_z = 0$, contrary to $L \in \mathrm{GL}_3$.

For type $(2, (2, 2))$, the ramification divisor has either two distinct roots or one double root. In the first case every proper parametrization is equivalent to

$$h_m = (x^4 + mx^3y, x^2y^2, y^4 + xy^3).$$

The exceptional values $m = 4$ and $m = 16$ are respectively the tricuspidal stratum and the previously excluded $(1, 3)$ stratum. Away from them, explicit quadratic syzygies a_m, b_m form a Hilbert–Burch basis. If $V = \alpha a_m + \beta b_m$, evaluation of (A.2) at the two ramification points gives

$$-216\beta^2x^6, \quad -24m(4\alpha + (m - 16)\beta)^2y^6.$$

They kill α, β ; the endpoint $m = 0$ is killed by the next three coefficient equations without dividing by m .

In the double-root case every proper parametrization is equivalent to

$$h_{A,B} = (x^4 + Ax^3y + Bx^2y^2, xy^3, y^4).$$

The $(2, 2)$ condition is precisely $3A^2 - 8B \neq 0$. For an explicit quadratic Hilbert–Burch basis a, b , the first two coefficients of $Q_h(\alpha a + \beta b)$ are

$$-12288\alpha^2x^6, \quad -6144\alpha(2A\alpha + \beta)x^5.$$

Thus $\alpha = 0$. The following three equations yield

$$u_2 = 32\beta^2, \quad u_1 = -8A\beta^2, \quad (8B - 3A^2)\beta^2 = 0,$$

and hence $\beta = 0$. In both root types the quadratic layer is then binary and the terminal equation again forces $L_z = 0$.

All displayed identities, the exceptional-parameter identifications, and the birational inverse formulas are checked both in SymPy and in an independent sparse-polynomial implementation over $\mathbb{Q}$. $\square$

Corollary A.2 (Exact leading target span in degree four). *Subject to the leading-curve reductions proved earlier in this manuscript, every nonautomorphic quartic Keller map has leading target span exactly two.*

Proof. The rank-one case is theorem 1.1. A full-span leading curve is linearly nondegenerate in $\mathbb{P}^2$. The conic and rational-cubic cases are excluded by the earlier leading-curve arguments, while theorem A.1 excludes the final proper rational-quartic cases. $\square$

A.2. An independent fixed-component check. The next calculation belongs to the rank-one route, not to the target-span-two theorem. We retain it because it closes the sole fixed-component chart left by the earlier $2+2$ calculation and supplies an independently implemented coefficient proof of that special case.

Proposition A.3 (Fixed-component Plücker boundary). *In the normalized chart*

$$H_4 = (0, 0, x^2y^2), \quad (H_3)_1 = x^2y, \quad (H_3)_2 = xy^2,$$

the Keller equations force the third column of L to vanish. Hence this chart cannot occur for a Keller map with invertible linear part.

Computer-assisted coefficient proof. The complete high-syzygy solution has parameters $t_1, t_2, u_1, \dots, u_4, c$. Selected degree-five coefficients first give

$$t_1 = t_2 = u_2 = u_3 = 0, \quad 9c + u_1^2 - 4u_1u_4 + u_4^2 = 0.$$

The xyz^2 -coefficient of D_4 and the z^3 -coefficient of D_3 then give

$$(u_1 - 2u_4)(u_1 + u_4)(2u_1 - u_4) = 0, \\ u_1u_4(u_1^2 - 4u_1u_4 + u_4^2) = 0.$$

Their only common zero is $u_1 = u_4 = 0$, so $c = 0$.

Writing the third column of L as $(\alpha, \beta, \gamma)^T$, two further coefficients determine the xz, yz terms of $(H_2)_3$ from α, β . Three square coefficients then give $\alpha = \beta = 0$, and the remaining degree-four coefficient is 3γ . Hence $\gamma = 0$. Both a SymPy construction of the full determinant and an independent standard-library sparse implementation verify this chain with every unrestricted lower coefficient present. $\square$

A.3. The target-span-two ramification filtration. In the binary-pencil locus write

$$H_4 = (P, Q, 0), \quad P, Q \in S_4, \quad R = (H_3)_3 \in S_3, \quad S = \mathbb{C}[x, y].$$

Set

$$U = J(Q, R), \quad V = J(P, R), \quad W = J(P, Q), \quad \Delta = \gcd(U, V, W).$$

The divisor Δ measures common ramification of the weighted binary map $[x : y] \mapsto [P : Q : R]$.

Theorem A.4 (Target-span-two exclusions through double ramification). *Assume proposition 6.1, the low-degree plane theorem, the cubic-cube coordinate reduction, and the primitive fourth-power exclusion used in this manuscript.*

- (i) *If the binary pencil is regular, $\Delta = 1$, then F is a polynomial automorphism.*
- (ii) *No quartic Keller map lies in the genuinely nonbinary, no-fixed-component quadratic-source locus.*
- (iii) *If P, Q are coprime and $\deg \Delta \leq 2$, then F is a polynomial automorphism.*

Consequently a nonautomorphic map in the coprime binary locus must satisfy

$$\deg \Delta \geq 3.$$

Proof. For (i), the maximal minors $(U, -V, W)$ of the two-row gradient matrix have height two. Hilbert–Burch gives two minimal syzygies of total degree eight. The coefficients of D_7 have total degrees five, six, and seven, so every z -dependent normal term in $(H_3)_1, (H_3)_2, (H_2)_3$ vanishes. The equation $D_6 = 0$ similarly kills the remaining nonlinear z -terms and the relevant entry of L . The map is triangular over a plane Keller map of degree at most four.

For (ii), write

$$P = U_0(a, b), \quad Q = V_0(a, b)$$

with a, b coprime quadrics and $[U_0 : V_0]$ of degree two. A valuation argument on the generic conic of the pencil shows that a nonbinary survivor must have a double-line fiber $a = x^2$ and

$$R = xb.$$

There are three nonbinary conic-pencil normal forms

$$b = x^2 + yz, \quad b = yz, \quad b = y^2 + xz,$$

and three relative normal forms for the outer double cover. Exact elimination in all nine charts, with arbitrary H_3, H_2, L , makes columns two and three of L proportional.

It remains to prove (iii) for $\deg \Delta = 1, 2$. At a simple ramification point, a source shear gives

$$\begin{aligned} P &= p_0x^4 + p_2x^2y^2 + p_3xy^3 + p_4y^4, \\ Q &= q_0x^4 + q_2x^2y^2 + q_3xy^3 + q_4y^4, \\ R &= r_0x^3 + r_2xy^2 + r_3y^3. \end{aligned}$$

The first residual normal parameter is supported on the vanishing of the weighted-inflection determinant

$$\Omega = \det \begin{pmatrix} 4p_0 & 4q_0 & 3r_0 \\ p_2 & q_2 & r_2 \\ p_3 & q_3 & r_3 \end{pmatrix}.$$

Classifying $\Omega = 0$ leaves two even families and one special orbit. The special orbit has the unavoidable coefficient

$$[xy^4z]D_6 = [y^5z]D_6 = \frac{104}{3}.$$

In each even family the remaining equations make $\text{col}_2(L)$ proportional to $\text{col}_3(L)$. Thus simple ramification is impossible.

For $\deg \Delta = 2$, the primitive Hilbert–Burch shifts are

$$(3, 7), \quad (4, 6), \quad (5, 5).$$

The first type enters the cubic-cube/fourth-power boundary. In type $(4, 6)$, a constant source vector field reduces the nonboundary chart to

$$R_y = \Delta, \quad P_y = x\Delta, \quad Q_y = y\Delta.$$

The four relative orbits of the quadratic divisor are checked exactly. They are inconsistent, acquire a fixed component or extra ramification, or leave the parameter-free obstruction

$$[y^3 z^2]D_5 = -\frac{1}{3}.$$

In type $(5, 5)$, a double root of Δ gives a fixed component. For squarefree Δ , the normalized forms are

$$\begin{aligned} P &= p_0 x^4 + p_2 x^2 y^2 + p_4 y^4, \\ Q &= q_0 x^4 + q_2 x^2 y^2 + q_4 y^4, \\ R &= r_0 x^3 + r_3 y^3. \end{aligned}$$

The two extreme coefficients of D_6 show that every nonzero normal direction produces a fourth-power pencil member. If the normal direction vanishes, lower syzygy degrees make the map triangular. This exhausts $\deg \Delta = 2$. $\square$

A.4. What remains in degree four. The next binary stratum is $\deg \Delta = 3$. Its primitive Hilbert–Burch types are $(2, 5)$ and $(3, 4)$. The first again enters the cubic-cube/fourth-power boundary. In the generic $(3, 4)$ chart, the first transverse normal branch has exact determinant

$$9216\tau^3(\tau^2 + 2\tau + 3)(3\tau^2 + 2\tau + 1);$$

the exceptional kernels have dependent P, Q , a cubic cube, or a common factor. A later calculation reduces the remaining generic normal deformations to an explicit weighted-inflection family F_4 .

An independent exact verifier over $\mathbb{Q}(a, b, \tau)$ reconstructs the generic $(3, 4)$ Hilbert–Burch chart, checks the gradient factorization and determinantal normal-form identities, and proves that the six next Keller equations have generic rank six and a two-dimensional kernel with the two recorded basis vectors. Over the algebraic closure, intersecting that kernel plane with the quadratic Veronese image forces the generic transverse parameters to vanish away from the displayed denominator, rank-drop, and weighted-inflection divisors. This exact calculation narrows the remaining work but does not check the exceptional divisors or the later F_4 compatibility problem.

We do not promote the complete $\deg \Delta = 3$ exclusion to a theorem. The conversation calculation leaves one precise compatibility check: after solving all of $D_6 = 0$ over

$$\mathbb{Q}(\tau)[d]/(q_4(d, \tau)),$$

one must prove uniformly that arbitrary lower binary terms cannot cancel the displayed D_5 obstruction on F_4 . The pure obstruction and exact sample parameters were checked, but no auditable attached program completes this extension-field solve.

The continuation in appendix B closes $\deg \Delta = 4, 5$, the zero-minor boundary, the primitive fourth-power boundary, and the fixed-component leaves, subject to the upstream routing used here. It leaves the exceptional

$\deg \Delta = 3$ F_4 system and a global exhaustiveness audit. This remains a reduction of the quartic problem, not a proof that ordinary degree four is impossible.

APPENDIX B. HIGH RAMIFICATION AND FIXED COMPONENTS

This appendix continues the target-span-two filtration of appendix A. Its conclusions are conditional on the leading-curve and four-loci reductions stated in the main text, and on the low-degree plane and cubic-coordinate theorems used there. The fixed-component argument also uses the standard fact that the polynomial Jacobian centralizer of an indecomposable polynomial in two variables is its polynomial ring. Full proof notes, finite chart lists, and exact programs are preserved in

`code/program-2-2026-07-29-v2/`.

Every supplied program was source-reviewed and replayed successfully. This verifies the encoded identities and finite exceptional charts; it is not an independent proof that the upstream geometric reductions are exhaustive.

B.1. Primitive high ramification and edge cases. Let

$$H_4 = (P, Q, 0), \quad P, Q \in k[x, y]_4, \quad \gcd(P, Q) = 1,$$

and put $R = (H_3)_3 \in k[x, y]_3$. Define

$$U = J(Q, R), \quad V = J(P, R), \quad W = J(P, Q),$$

and, when all three minors are nonzero,

$$H = \gcd(U, V, W), \quad r = \deg H.$$

Theorem B.1 (Primitive binary high ramification). *Assume the Program 2 reductions place a quartic Keller map in the primitive coprime binary-pencil locus above. If $r = 4$ or $r = 5$, then the map is a polynomial automorphism.*

Proof. Write $U = Hu, V = Hv, W = Hw$. The planar minors satisfy

$$u \, dP - v \, dQ + w \, dR = 0, \tag{A.1}$$

with

$$\deg u = \deg v = 5 - r, \quad \deg w = 6 - r.$$

For $r = 5$, u, v are constants. If $S = uP - vQ$, then

$$dS = -w \, dR, \quad dw \wedge dR = 0,$$

so homogeneity gives

$$R = cw^3, \quad S = -\frac{3c}{4}w^4.$$

The resulting aligned cube-fourth-power pair is automorphic by the cubic-coordinate and plane-degree reduction.

For $r = 4$, first suppose u, v are dependent. After a target change,

$$\ell \, dS + w \, dR = 0, \quad 4\ell S + 3wR = 0. \tag{A.2}$$

If $\ell \mid w$, equation (A.2) again produces an aligned cube–fourth-power pair. If $\ell \nmid w$, it forces $\ell^4 T = cw^3$ with $R = \ell T$, which is incompatible with unique factorization.

Now suppose u, v are independent and choose $u = x, v = y$. Euler contraction and differentiation of (A.1) give

$$P = \frac{wR_x - 3Rw_x}{4}, \quad Q = \frac{3Rw_y - wR_y}{4}. \quad (\text{A.3})$$

If the residual quadratic w is a square, (A.3) makes P, Q share a linear factor, contrary to primitivity. If w is reduced, the determinant arc reduces the first transverse normal direction to two binary linear forms ℓ, m . The necessary divisibility is

$$\Gamma \mid \frac{1}{2}(\mathcal{A}\ell^2 + 2\mathcal{B}\ell m + \mathcal{C}m^2), \quad \mathcal{A}\mathcal{C} - \mathcal{B}^2 = 4xy\Gamma. \quad (\text{A.4})$$

For squarefree Γ , reduction modulo Γ gives $\ell = m = 0$. The repeated-root incidence has two components: one violates $\gcd(P, Q) = 1$, and the other is a rational conic. The exact homogeneous parametrization checks its affine chart, its omitted point at infinity, and the $2 + 2$ divisor, where the complete first-normal kernel is two dimensional. In each chart the next determinant coefficient kills the kernel. A later coefficient then kills the remaining quadratic normal directions. All nonlinear terms are binary, so a target change reduces the map to a plane Keller pair with one coordinate of degree at most four. $\square$

Proposition B.2 (Fourth-power and zero-minor boundaries). *Subject to the same structural inputs:*

- (i) *if a primitive coprime leading pencil contains a fourth power, then its cubic and quadratic normal layers route it to the binary, quadratic-source, or aligned pure-power branch, so it creates no additional open leaf;*
- (ii) *if one of U, V, W vanishes, including the separate case $R = 0$, then the map is a polynomial automorphism.*

Proof. For (i), normalize $H_4 = (x^4, Q, 0)$, $x \nmid Q$. The highest determinant equation makes $Q(1, u, v)$ and $R(1, u, v)$ algebraically dependent. Their common polynomial generator has degree one or two, routing respectively to the binary or quadratic-source branch; a constant generator gives an aligned pure power. If $R = 0$, the same argument applies to $E = (H_2)_3$.

For (ii), if $J(Q, R) = 0$ with $Q, R \neq 0$, Euler’s identity gives

$$3R dQ - 4Q dR = 0,$$

and hence $Q = \alpha L^4, R = \beta L^3$. The other one-minor case is symmetric, and two vanishing minors with $R \neq 0$ contradict target span two. When $R = 0$, the next determinant equation is

$$J(P, Q)E_z = 0,$$

so $F_3 = e(x, y) + \mu z$, $\deg e \leq 2$. Straightening this coordinate reduces the remaining two coordinates to a plane Keller map with a coordinate of degree at most seven; the corrected low-degree plane theorem applies. $\square$

B.2. The corrected fixed-component valuation. Write

$$P = GA, \quad Q = GB, \quad \gcd(A, B) = 1,$$

where $g = \deg G$, and again put $R = (H_3)_3$. In this primitive branch define

$$t = A/B, \quad \frac{R^4}{Q^3} = \rho(t).$$

Lemma B.3 (Fixed-component valuation). *Let an irreducible component Γ occur in G with multiplicity s and in the reduced special fiber $A - \xi B$ with multiplicity m . Put*

$$c_\xi = \text{ord}_\xi \rho \quad (\xi \neq \infty), \quad c_\infty = \text{ord}_\infty \rho + 3.$$

Then

$$4\nu_\Gamma(R) = 3s + c_\xi m, \quad \sum_{\xi \in \mathbb{P}^1} c_\xi = 3. \quad (\text{A.5})$$

For a nonfixed component in the same fiber, equation (A.5) holds with $s = 0$.

Proof. At a finite fiber, B is a unit along Γ , and $\nu_\Gamma(t - \xi) = m$. Taking the valuation of $R^4/Q^3 = \rho(t)$ gives the first identity. At infinity, B has order m , producing the added 3 in the definition of c_∞ . The second identity is the degree-zero divisor identity for ρ , with that adjustment. $\square$

The term $3s$ is essential, and the numbers c_ξ need not be nonnegative: a pole of ρ can be cancelled by G^3 . Thus any earlier provisional closure that omitted $3s$ or imposed $c_\xi \geq 0$ at every fixed fiber is superseded by lemma B.3.

Theorem B.4 (Genuinely nonbinary fixed components). *Assume the four-loci reduction and the polynomial-centralizer fact stated at the beginning of this appendix. Every genuinely nonbinary primitive fixed-component branch is closed.*

Proof. If $\deg G = 2$, equations (A.5) for a fixed line and the residual non-fixed line give incompatible congruences modulo four, unless the whole reduced fiber is fixed; that endpoint is the already-recorded same-fiber rank-one chart.

If $\deg G = 1$, let $m \in \{1, 2, 3\}$ be the multiplicity of the fixed line in its reduced cubic fiber. Multiplicity one conflicts with the valuation on the residual factor, and multiplicity two is impossible modulo four. For multiplicity three, the only possibilities are the aligned automorphic case or

$$H_4 = (x^4, xR, 0), \quad (H_3)_3 = R, \quad x \nmid R. \quad (\text{A.6})$$

The binary common-generator subcase stays in the binary tree. In the genuinely nonbinary subcase, the Hamiltonian derivation $\delta = J_{y,z}(R, -)$ has kernel $k(x)[R]$. The first two normal equations give, after target changes,

$$\begin{aligned} A &= ax^3, & B &= xE + \frac{a}{4}R - \frac{\gamma}{4}x^3, \\ C &= \mu x^2, & D &= xL_3 + \frac{a}{4}E - \frac{\nu}{4}x^2. \end{aligned} \tag{A.7}$$

Reassembly yields

$$F_1 = f(x) + \ell, \quad F_2 = (x + a/4)F_3 + p(x) + m.$$

The homogeneous degree-five determinant equation is

$$4x^3 \det(dx, dm, dR) - R \det(dx, d\ell, dR) = 0. \tag{A.8}$$

Since $x \nmid R$, both determinants in (A.8) vanish. Nonbinary primitivity then forces $\ell, m \in kx$, making two differential rows dependent and contradicting the Keller condition. $\square$

B.3. Binary fixed factors.

Theorem B.5 (Binary fixed-factor exclusion). *Assume the Program 2 reductions reach a binary leading pencil*

$$H_4 = (P, Q, 0)$$

with a nonconstant greatest common divisor G . If $\deg G = 1, 2$, or 3 , then every quartic Keller map in the corresponding stratum is a polynomial automorphism.

Computer-assisted proof. For $\deg G = 3$, normalize $H_4 = (xG, yG, 0)$. The three multiplicity types of G are squarefree, double-plus-simple, and a triple line. On the squarefree chart, the first-normal determinant is

$$-6^6 \operatorname{Disc}(G) \operatorname{Res}(G, R)^2.$$

The exact root-incidence sweep closes its resultant divisor. Separate exact sweeps of the double-line and triple-line types either kill the normal amplitudes or give a triangular degree drop.

For $\deg G = 1$, the valuation reduction normalizes

$$H_4 = (x^4, xB, 0), \quad x \nmid B.$$

There are two affine orbits for the residual cubic:

$$B = y^3, \quad B = y^3 + x^2y.$$

In general coordinates the first-normal determinant is

$$248832 b_3^3 r_3^4 \Theta,$$

where

$$\begin{aligned} \Theta &= 3b_1^2 r_3^2 - 4b_1 b_2 r_2 r_3 - 6b_1 b_3 r_1 r_3 + 4b_1 b_3 r_2^2 \\ &\quad + 4b_2^2 r_1 r_3 - 4b_2 b_3 r_1 r_2 + 3b_3^2 r_1^2. \end{aligned}$$

The exact exceptional-divisor sweep treats the projective endpoints and the resonant vertex; the latter reassembles to a plane reduction.

For $\deg G = 2$, the coprime residual quadratic pencil is a secant of the Veronese conic and can be normalized as

$$P = x^2G, \quad Q = y^2G.$$

Writing

$$G = g_0x^2 + g_1xy + g_2y^2, \quad R = r_0x^3 + r_1x^2y + r_2xy^2 + r_3y^3,$$

the 8-by-8 first-normal determinant factors exactly as

$$\begin{aligned} & 41472(4g_0g_2 - g_1^2)(-4g_0r_1 + 3g_1r_0) \\ & \cdot (-3g_1r_3 + 4g_2r_2) \operatorname{Res}(G, R)^2. \end{aligned} \tag{A.9}$$

The four divisors in (A.9), all their intersections, rank jumps, projective endpoints, and the final aligned resonance are covered by 38 exact replay groups. No continuity argument across an inverted factor is used. Off the divisors the normal matrix is invertible; on every divisor the determinant arc either produces a unit obstruction or reduces to a plane automorphism. $\square$

Corollary B.6 (Revised quartic frontier). *Conditional on the complete upstream routing into the recorded Program 2 charts, the only unresolved degree-four calculation is the exceptional triple-ramification F_4 compatibility system. A separate global case-tree audit must still verify that the leading-curve and four-loci reductions route every quartic map into exactly one recorded terminal chart.*

This corollary is not a proof that ordinary degree four is impossible and does not establish $D_{\min} \geq 5$. Independent specialist proof review and reproduction of the new exact calculations also remain publication requirements.

APPENDIX C. FIXED-FACTOR CONIC STRATA IN DEGREES FIVE AND SIX

This appendix pushes the conic leading-image method beyond degree four. It does not classify degrees five or six: the theorem isolates an open basepoint boundary and a resonant primitive sextic case.

Let the leading homogeneous map have the squarefree fixed-factor form

$$\begin{aligned} H_D &= G(A^2, AB, B^2), \\ \deg G &= g > 0, \quad \deg A = \deg B = e, \quad D = g + 2e. \end{aligned}$$

Assume that A, B are a regular sequence and that some irreducible component of $G = 0$ sees a nonconstant ratio A/B . The latter is the basepoint-free invariant-gap locus used below.

Lemma C.1 (Invariant-degree gap). *For*

$$\Omega = A dG \wedge dB - B dG \wedge dA + 2G dA \wedge dB,$$

let δ be the associated homogeneous derivation. Every nonzero homogeneous P satisfying $\delta(P) = 0$ has degree divisible by D .

Proof. Choose an irreducible factor $\Gamma \mid G$, of degree h , on which A/B is nonconstant, and write $P = \Gamma^r Q$ with $\Gamma \nmid Q$. On the function field of $\Gamma = 0$, both the induced derivation and the Euler derivation annihilate A/B ; hence $\bar{\delta} = cE$ for a nonzero rational function c . From $\delta(GB^2) = 0$ one gets

$$\overline{\delta\Gamma/\Gamma} = -c(D - h).$$

Reducing $\delta(\Gamma^r Q) = 0$ modulo Γ now yields

$$c(\deg P - rD)\bar{Q} = 0.$$

Thus $\deg P = rD$. □

Write a Keller jet as

$$K_\varepsilon = H_D + \varepsilon H_{D-1} + \cdots + \varepsilon^{D-2} H_2 + \varepsilon^{D-1} L X$$

and set

$$\Phi(Y) = Y_1 Y_3 - Y_2^2, \quad \Phi(K_\varepsilon) = \sum_{j \geq 1} \varepsilon^j S_j.$$

The chain-rule identity

$$\text{Jac}(K_{\varepsilon,1}, K_{\varepsilon,2}, \Phi(K_\varepsilon)) = K_{\varepsilon,1} \det JK_\varepsilon$$

and lemma C.1 imply inductively that

$$S_1 = \cdots = S_{D-1} = 0.$$

Indeed, $\deg S_j = 2D - j$, which lies strictly between D and $2D$.

Theorem C.2 (Basepoint-free fixed-factor conic exclusion). *Under the preceding hypotheses, no nonautomorphic Keller map has one of the following leading-form types:*

$$\begin{array}{l|l} D = 5 & (g, e) = (1, 2), (3, 1) \\ D = 6 & (g, e) = (2, 2), (4, 1). \end{array}$$

Proof idea. The vanishings $S_1, \dots, S_{D-1}$ combine with the Koszul syzygies of (A, B) and the Hilbert–Burch syzygies of $(A, B)^2$. They lift the deformed rank-one symmetric matrix to

$$\widehat{M}_\varepsilon = a_\varepsilon P_\varepsilon P_\varepsilon^T + \varepsilon^3 \text{sym}(P_\varepsilon, Q_\varepsilon)$$

after a source-independent target correction. Consequently

$$\Phi(\widehat{K}_\varepsilon) = -\varepsilon^6 (P_\varepsilon \wedge Q_\varepsilon)^2.$$

The chain-rule identity forces every non- ε irreducible factor of $P_\varepsilon \wedge Q_\varepsilon$ to divide both of the first two coordinates and then the Keller determinant. This is impossible. The remaining possibility is a pure power of ε ; comparison of the lowest coefficients makes the wedge vanish and again forces the linear determinant to be zero.

Under `code/degree-five-six/`, the file `conic_fixed_factor_d5_d6_proof_note.md` writes out the four graded lift identities. The script `verify_conic_hensel_d5_d6.py` replays them exactly. □

The $(g, e) = (3, 1)$ quintic case admits a stronger boundary analysis.

Theorem C.3 (Three-variable cubic-factor quintics). *Let G be a squarefree cubic genuinely depending on x, y, z . There is no Keller jet with leading form*

$$G(x, y, z)(x^2, xy, y^2).$$

Thus the only unresolved cubic-factor overlap of this shape is the binary case $G \in \mathbb{C}[x, y]$.

Proof structure. At the base point $[0 : 0 : 1]$, the four conic normal equations split according to the multiplicity of $G = 0$ and whether the first normal linear form vanishes there. The unit branch is eliminated by the rank of the tangent cone and, in the rank-one case, by two incompatible coefficients. In the regular branch a source translation removes the first syzygy. The remaining cusp, node, transverse line-conic, and tangent line-conic normal forms each collapse under two or three displayed coefficients of the Keller determinant. Multiplicity three forces G to be binary.

The complete normal-form ledger and decisive coefficients are recorded in `quintic_cubic_factor_basepoints.md`, under `code/degree-five-six/`. The companion exact script replays every coefficient identity used in the case table. $\square$

Question C.4 (Binary cubic-factor overlap). *Can a quintic Keller map have leading form*

$$G(x, y)(x^2, xy, y^2)$$

for a squarefree binary cubic G ?

Question C.5 (Fixed-factor basepoints in degree five). *Can the remaining $(g, e) = (1, 2)$ quintic branches, where the base locus of (A, B) meets $G = 0$, satisfy the Keller equations?*

Question C.6 (Primitive sextic conic stratum). *Can a degree-six Keller map have primitive conic leading form*

$$(A_3^2, A_3B_3, B_3^2)$$

with no nonconstant fixed factor? Here degree three is resonant, so the invariant-gap argument does not apply.

APPENDIX D. BOUNDED-DEGREE SCHEMES AND THE SHARP EQUIVARIANT THRESHOLD

This appendix fixes the finite-algebra framework. It also records the exact classification for weights $(-1, 1, 2)$. These calculations originally motivated the unrestricted quartic analysis in the body of the paper.

D.1. The coefficient scheme and collision slice. Fix n, D . After translating the source and target and normalizing the linear part, write

$$F = X + \sum_{d=2}^D H_d.$$

Equating the nonconstant coefficients of $\det JF - 1$ defines a finite-type coefficient scheme $K_{n,D}$. At a Keller map F , its tangent equation in the direction H is

$$\mathrm{tr}((JF)^{-1}JH) = 0$$

coefficientwise.

If a Keller map has a collision, affine source and target transformations place the two source points at $0, e_1$, their common image at 0 , and the first Jacobian at the identity, without increasing ordinary degree. Thus a fixed-degree counterexample search can be conducted in the finite closed incidence slice

$$F(0) = F(e_1) = 0, \quad JF(0) = I, \quad \det JF = 1.$$

This is an implementation framework rather than a novelty claim about finite coefficient loci themselves.

D.2. Equivariant quotient equation. Let

$$\lambda \cdot (x, y, z) = (\lambda^{-1}x, \lambda y, \lambda^2 z), \quad u = xy, \quad v = x^2 z.$$

Every normalized equivariant map has a unique Laurent presentation

$$X = xf(u, v), \quad Y = x^{-1}g(u, v), \quad Z = x^{-2}h(u, v).$$

Polynomiality imposes finite support conditions on f, g, h . The Keller equation is the planar identity

$$gJ(f, h) - 2hJ(f, g) + fJ(g, h) = 1, \tag{D.1}$$

where J denotes the Jacobian in u, v .

Lemma D.1 (Boundary equation). *Let $C_f = V(f)$. Then C_f is smooth and*

$$gD_f h - 2hD_f g = 1 \quad \text{on } C_f, \quad D_f = J(f, -).$$

On an affine-line component, if this is written

$$g_0 h'_0 - 2h_0 g'_0 = \kappa \neq 0,$$

then $\deg g_0 \leq 1$.

Proof. At a singular point of $f = 0$, the left side of (D.1) would vanish. For the degree assertion, compare the leading term. If cancellation occurs because $\deg h_0 = 2 \deg g_0$, subtract a multiple of g_0^2 from h_0 ; this preserves the equation and lowers the degree. Iteration leaves $\deg g_0 \leq 1$. $\square$

Theorem D.2 (Exact equivariant threshold). *Every normalized $(-1, 1, 2)$ -equivariant Keller map of ordinary degree at most six is a polynomial automorphism. The known degree-seven counterexample has this symmetry. Hence the exact minimum in this equivariant class is seven.*

Proof and certificate structure. For degree at most five, the exhaustive support is

$$\begin{aligned} f &= 1 + au + bv + cu^2, \\ g &= u + pv + qu^2 + ruv + su^3 + tv^2, \\ h &= v + Au^2 + Buv + Cu^3 + Dv^2 + Eu^2v. \end{aligned}$$

Lemma D.1 reduces the nonconstant- f locus to four exact coefficient ideals. After the necessary saturations, their Gröbner bases over $\mathbb{Q}$ are all (1). Thus $f = 1$, and (D.1) reduces to $J(g, h) = 1$ for a plane map of degree at most three. It follows that the three-dimensional map is birational and hence an automorphism.

For degree at most six, the exhaustive support is

$$\begin{aligned} f &= 1 + au + bv + cu^2 + duv, \\ g &= u + pv + qu^2 + ruv + su^3 + tv^2 + ku^2v, \\ h &= v + Au^2 + Buv + Cu^3 + Dv^2 + Eu^2v + Fu^4 + Guv^2. \end{aligned}$$

When $d \neq 0$, normalize $d = 1$ and $b \in \{0, 1\}$. Smoothness is the nonvanishing of

$$\Delta = 1 - ab + cb^2,$$

and both ideals saturated by Δ are unit ideals over $\mathbb{Q}$.

When $d = 0, b \neq 0$, normalize $b = 1$. The boundary lemma gives the necessary relations among q, r, s, t, k ; the five resulting charts, including the previously easy-to-miss $c = 0, k \neq 0$ chart, all have unit Gröbner ideals.

Finally suppose $b = d = 0$, so $f = f(u)$. If f is nonconstant, write

$$g = n(u) + m(u)v, \quad h = L(u) + j(u)v + N(u)v^2.$$

With

$$U = fg, \quad V = f^2h,$$

equation (D.1) becomes

$$J(U, V) = f^2.$$

Eliminating v shows that the U^2 - and U -coefficients of V must be constant and that

$$\left. \frac{dV}{du} \right|_U = -\frac{f}{m}.$$

The cases m nonconstant, m a nonzero constant, and $m = 0$ each force the impossible divisibility $f \mid u$. Therefore $f = 1$, and the remaining plane map has degree at most four and is an automorphism.

The finite branches and unit-ideal certificates were generated over $\mathbb{Q}$ and rerun from the recovered exact scripts. Finite-field scans were useful diagnostics but are not logical inputs to the theorem. $\square$

Remark D.3 (A corrected boundary warning). The Laurent boundary equation alone does not force a monomial solution. There are explicit nonmonomial Laurent pairs satisfying it. The proof above uses the complete lifted Keller identity, not that false intermediate shortcut.

D.3. The unrestricted quartic coefficient problem. For normalized maps $\mathbb{A}^3 \rightarrow \mathbb{A}^3$ of ordinary degree at most four, the unrestricted collision coefficient system has 93 variables and 219 nonconstant determinant-coefficient equations. This count is exact, but it does not by itself make the system tractable.

Inside the weight- $(1, -1, -2)$ quartic subfamily, exact coefficient elimination makes every Keller map tame. The calculation is a useful negative result for that symmetry class; it is not a classification of unrestricted quartic Keller maps. In particular, no lower bound for a general counterexample follows merely from this computation.

Question D.4 (Degrees five and six). *Can the unrestricted degree-five and degree-six collision schemes be reduced to finitely many invariant leading-form strata with certificates comparable to those above? More specifically, which parts of the leading-curve, maximal-contact, and Keller-jet analysis used in degree four persist before the degree-seven example appears?*

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AI systems were used extensively for exploration, proof drafting, symbolic code, literature triage, and manuscript editing. They are not authors. The human author is responsible for every mathematical claim. Exact symbolic checks are identified in section 9; they do not replace review of the geometric reductions.

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